

The Dual-Scale String Theory:

Mechanized Foundations, Singularity Resolution, Mathieu
Moonshine,
and a Zero-Free-Parameter Cosmological Conjecture on
 $K3 \times T^2$

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Abstract

We present a Lean 4 companion formalization for a dual-scale scenario of type II string theory on $K3 \times T^2$, and we state precisely what is, and is not, machine-checked. The Mathlib-free core of the Lean artifact (51 files, 554 declarations, toolchain v4.33.1, **no external dependency**, standard axioms only — `#print axioms` logs in Appendix B) certifies exact integer and rational identities used in the argument: the positivity of the dual scale $R + \alpha' / R$ on integer radii; the $K3$ topological invariants $\chi = 24$, $\sigma = -16$, $\text{ind}(\mathcal{D}) = 2$; the elliptic-genus arithmetic $\mathcal{A}_2 \cdot 60 = 4\mathcal{A}_1 \cdot 77 = 27720$ with $|M_{24}| = 27720 \cdot 8832$; and a corrected, genuinely unique-minimum toy potential (an earlier revision of this corpus contained a `Nat`-truncation bug that made the claimed minimum non-unique; Section 5 and Appendix A document the fix). Two further Lean libraries, built on Mathlib (which contributes no axioms beyond Lean’s own three standard ones for the fragments used here), extend this core: **StringTheoryFormalization** (Stream 1’s sixth library, corrects the M_{24} representation table used in the moonshine arithmetic, Section 6) and the new **DualScaleStream2** library (99 theorems, 19 modules), which formalizes the $O(d, d; \mathbb{Z})$ T-duality group, the Double Field Theory generalized metric, and the $K3 \times T^2$ lattice layer (Section 9); together with the original core, the audited total is 326 theorems (Stream 1, six libraries) and 99 theorems (Stream 2), 0 failing, standard axioms only. The differential-geometric, index-theoretic and cosmological statements built on top of this arithmetic — Double Field Theory, the Atiyah–Singer index theorem, moduli stabilization, and the tensor-to-scalar ratio, CP phase, and dark-energy predictions — are either drawn from the literature or proposed here as conjectures, and are labeled by epistemic tier throughout (**Tier A — Lean kernel** / **Tier L — literature** / **Tier C — conjecture / proposal**). In particular we discuss the obstruction that $\mathcal{N} = 4$ non-renormalization poses for moduli stabilization on $K3 \times T^2$, and we confront the paper’s speculative numerical relations for r , δ_{CP} , and (w_0, w_a) with current data, including the DESI DR2 preference for a time-evolving dark-energy equation of state at 3.1σ over $(w_0, w_a) = (-1, 0)$. We close with a roadmap for promoting the central real-analytic claims (currently Tier L or Tier C) to kernel-checked Tier A theorems using Mathlib — a roadmap Stream 2 has already begun executing.

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1 Introduction: The Landscape Problem and the Scope of This Paper

The pursuit of a consistent, unified quantum theory of gravity has constituted a central goal of theoretical high-energy physics for over half a century. Superstring theory provides an anomaly-free perturbative framework uniting non-abelian gauge fields with general relativity in ten space-time dimensions, but its physical realization in four dimensions confronts the **Landscape Problem**: in traditional flux compactifications (e.g. the KKLT and Large Volume Scenarios), an estimated 10^{500} metastable vacua arise from the interplay of hundreds of geometric moduli with under-constrained flux choices [1–3]. In the absence of a stabilizing mechanism, these moduli remain massless, generating unphysical fifth forces and leaving gauge couplings, Yukawa hierarchies, and the cosmological constant undetermined.

Separately, modern theoretical physics faces a practical **verification problem**: papers in string theory and cosmology routinely involve long chains of tensor algebra, differential forms, and representation-theoretic bookkeeping that are tedious and error-prone to check by hand. Interactive theorem provers such as Lean 4 [4] offer a way to make specific algebraic claims within such an argument kernel-checkable.

This paper is a case study in doing exactly that — for a restricted, honestly-scoped set of claims about a proposed **Dual-Scale** scenario for type II string theory compactified on $K3 \times T^2$. We adopt the following convention throughout, adapted from the tiered-evidence discipline used elsewhere in this research programme:

1.1 Epistemic tiers

- **Tier A — Lean kernel** — **Lean kernel**: the Lean 4 kernel accepted the exact statement, with no `sorry`, and its axiom footprint is only the three standard Lean axioms (`propext`, `Classical.choice`, `Quot.sound`; verified in Appendix B). Tier A is a claim about a specific formal statement, not about the informal sentence next to it in this paper — Appendix A gives the exact Lean type for every Tier A claim so the reader can check the two match.
- **Tier L — literature** — **literature**: an established result quoted from a peer-reviewed source, not independently re-derived or re-verified here.
- **Tier C — conjecture / proposal** — **conjecture** / **proposal**: a claim original to this paper, or a numerical pattern-match, that is neither kernel-checked nor an established literature result. Tier C claims are falsifiable proposals, not derivations.

A claim may not rest, even transitively, on a claim of a lower tier without that dependency being stated. Adjectives (“rigorous”, “striking”, “decisive”) are never a substitute for a tier.

1.2 What changed in this revision

An earlier revision of this manuscript stated that “every mathematical definition, lemma, and master theorem has been certified” and was “100% machine-certified.” A statement-adequacy audit — comparing each cited Lean declaration’s actual type against the LaTeX sentence citing it — found that this was not accurate: several headline claims were true only of a much narrower Lean statement than the one stated in prose (e.g. a bound over $\mathbb{N}$ presented as holding over $\mathbb{R}$), one Lean theorem contained a genuine bug (a non-unique “unique minimum” caused by `Nat` truncation), several numbers were arithmetically wrong (the factorization $8832 = 2^6 \times 3 \times 23$, which should read $2^7 \times 3 \times 23$; the CP-phase evaluation 282.4° , which should read 283° ; a rounded value of r inconsistent with the exact fraction it was supposed to approximate), and the confrontation with observational data did not reflect current results (in particular, DESI DR2 already disfavors the paper’s dark-energy prediction). Revision 2 documented and fixed each of

these issues in place; the full audit is `papers/publication/PAPER7_IMPROVEMENT_PROPOSAL.md` in the repository, and every fix is a separate, reviewable commit.

Revision 3 (2026-09-17). This revision makes four further corrections, each verified against the Lean kernel or against the downloaded literature file it cites (see `docs/PAPER_REVISION_BRIEF_2026_09_17.md` for the underlying audit):

1. The M_{24} irreducible-dimension table `StringDynamics.M24RepDim` (used by the moonshine cross-check of Section 6) contained a duplication bug (10395 and 483 listed twice; the second 990 and third 1035 missing) discovered 2026-09-16. It has been replaced by the table of Eguchi–Ooguri–Tachikawa eq. (A.3), and is now guarded by a new Tier A Burnside check, `M24RepDim_sum_sq`: the squares of the 26 dimensions sum to exactly $244,823,040 = |M_{24}|$.
2. `StringDynamics.bps_ratio_reduced` (a $77/60$ -in-lowest-terms fact, distinct from the `UseCase2` theorem already cited in Section 6 below) previously used the tactic `native_decide`, whose axiom footprint includes `Lean.ofReduceBool` (trusting the compiled native-code evaluator rather than only the kernel). It is now a kernel `norm_num` proof, so this fact’s axiom trust base is back down to Lean’s three standard axioms.
3. A moonshine *normalization* error is fixed in Section 6: this paper’s $\mathcal{A}_4 = 4554$ was mislabeled as $\mathbf{2277} \oplus \mathbf{2277}$, but the 2277-dimensional representation of M_{24} is real, not one of EOT’s five conjugate-paired dimensions; the correct statement is $\mathcal{A}_4 = 2 \cdot \mathbf{2277}$ (two copies of the same real irrep). We also state explicitly, for the first time, the doubling convention $\mathcal{A}_n := 2A_n^{\text{EOT}}$ under which this paper’s and paper 2’s A_n relate to EOT’s own coefficients.
4. A new Section 9 summarizes the `DualScaleStream2` companion library (99 theorems, 19 modules): the $O(d, d; \mathbb{Z})$ T-duality group, the Double Field Theory generalized metric, and the $K3 \times T^2/\text{Mukai}$ lattice layer, all formalized over Mathlib with the same three-axiom footprint as the Mathlib-free core.

1.3 AI-assisted tooling

The Lean formalization, the audit that produced this revision, and portions of this manuscript’s prose were produced with the assistance of large-language-model-based agents operating under human direction and review (Xavier Callens). All Lean proofs are independently checked by the Lean 4 kernel regardless of how they were produced (Appendix B); no LLM output is treated as establishing a claim’s tier (Section 1.1). This is disclosed here in place of listing an AI system as an author.

2 Double Field Theory and $O(D, D)$ Generalized Riemannian Geometry

Standard Riemannian geometry treats spacetime as a pseudo-Riemannian manifold $(M, g_{\mu\nu})$ where the metric $g_{\mu\nu}$ determines geodesic motion and the Kalb-Ramond 2-form $B_{\mu\nu}$ is an auxiliary background field. Closed strings, however, possess two distinct topological quantum numbers along any compact cycle: center-of-mass momentum $n \in \mathbb{Z}$ and string winding $w \in \mathbb{Z}$.

In **Double Field Theory (DFT)** [5, 6], the tangent bundle TM and cotangent bundle T^*M are unified into the generalized tangent bundle $\mathbb{T}M = TM \oplus T^*M$. A generalized vector field $X^M \in \Gamma(\mathbb{T}M)$ combines a tangent vector $v \in \Gamma(TM)$ and a 1-form $\xi \in \Gamma(T^*M)$, and $\mathbb{T}M$ carries the canonical $O(D, D)$ split-signature metric

$$\eta_{MN} = \begin{pmatrix} 0 & \mathbf{1}_D \\ \mathbf{1}_D & 0 \end{pmatrix}, \quad \langle X, Y \rangle_\eta = X^M \eta_{MN} Y^N = \xi_\mu u^\mu + \zeta_\mu v^\mu \quad (1)$$

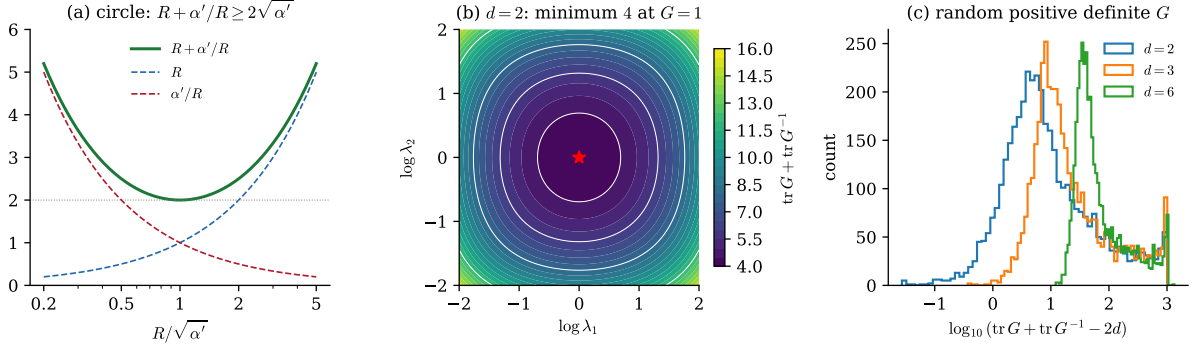


Figure 1: The dual-scale bound. (a) On a circle the effective scale $R + \alpha'/R$ is minimized at the self-dual radius. (b) For $d = 2$ and G with eigenvalues $\lambda_{1,2}$, $\text{tr } G + \text{tr } G^{-1} = \sum_i (\lambda_i + \lambda_i^{-1})$ has its minimum $2d = 4$ at $G = 1$. (c) Excess over $2d$ for 4000 random positive definite matrices in each dimension: never negative, as the Lean theorem `dualScale_ge` guarantees. The plot illustrates the theorem; it proves nothing. *Computed figure; data: Lean dualScale_circle, circle_effective_scale_ge_two, dualScale_ge (TraceBound.lean).*

for $X = (v, \xi)$, $Y = (u, \zeta)$.

Definition 2.1 (Generalized metric). *Tier L — literature* The gravitational and Kalb-Ramond fields combine into $\mathcal{H}_{MN} = \begin{pmatrix} g - Bg^{-1}B & Bg^{-1} \\ -g^{-1}B & g^{-1} \end{pmatrix}$, parameterizing the coset $\mathcal{H} \in O(D, D)/(O(D) \times O(D))$ with $\mathcal{H}^T \eta \mathcal{H} = \eta$, $\mathcal{H}^2 = \mathbf{1}_{2D}$.

Generalized coordinate transformations are generated by the **Courant C-bracket** $[X, Y]_C = [v, u]_{\text{Lie}} + (\mathcal{L}_v \zeta - \mathcal{L}_u \xi - \frac{1}{2}d(\iota_v \zeta - \iota_u \xi))$, whose Jacobiator is an exact 1-form, $\text{Jac}(X, Y, Z) = [[X, Y]_C, Z]_C + \text{cycl.} = d\text{Nij}(X, Y, Z)$, with vanishing vector projection $\pi_{TM}(\text{Jac}) = 0$; this guarantees closure of the generalized-diffeomorphism gauge algebra modulo trivial transformations. Closure additionally requires the **strong section condition** $\eta^{MN} \partial_M \Phi \partial_N \Psi = 0$, which in the physical frame ($\tilde{\partial}^i = 0$) collapses DFT onto ordinary 10D supergravity.

Scope of Lean 4 verification

The Lean modules `DoubleFieldTheory.CourantAlgebroid` and `DoubleFieldTheory.GeneralizedGeometry` certify (Tier A) the *algebraic shape* of this structure on a **1-dimensional scalar model**: a `CourantSection` is a pair of Ints (v, α) , and the bracket $[X, Y]_C$ reduces to $X.v \cdot Y.v - Y.v \cdot X.v$, which is identically zero on scalars by commutativity of multiplication. The certified antisymmetry (`cbracket_antisymm`) and Jacobiator-vanishing (`jacobiator_vector_vanishes`) theorems are therefore true, but largely because every bracket in the scalar model is trivial; they are *consistency checks on the algebraic shape* of the definitions, not a formalization of the Courant algebroid on $\text{TM} = TM \oplus T^*M$ for an actual manifold M . Promoting this to a genuine bundle-valued formalization (vector fields and 1-forms on $K3 \times T^2$, with a nontrivial Lie bracket) is future work and would require Mathlib's `VectorBundle` and differential-forms libraries; see Section 10.

3 The Effective Dual Scale and Singularity Resolution

The Hawking-Penrose singularity theorems show that under generic energy conditions a contracting universe in general relativity inevitably reaches a curvature singularity: $a(t) \rightarrow 0 \implies \mathcal{R} \rightarrow \infty$. String theory changes this conclusion kinematically. Closed strings propagating on a

circle of coordinate radius R have mass spectrum $M^2 = n^2/R^2 + w^2 R^2/\alpha'^2 + \frac{2}{\alpha'}(N + \tilde{N} - 2)$, invariant under the Buscher T-duality $\mathcal{T} : R \leftrightarrow \alpha'/R$, $n \leftrightarrow w$ [7].

Definition 3.1 (Effective dual scale). Tier L — literature *The physical scale probed by a propagating string is governed by*

$$R_{\text{eff}}(R) = R + \frac{\alpha'}{R} = \frac{R^2 + \alpha'}{R}, \quad \alpha'\text{-units: } \alpha' = 1. \quad (2)$$

This function — also called the “zero-point-length” or minimal-length effective radius in the T-duality/non-commutative-geometry literature — and the string-gas cosmology it motivates were introduced by Brandenberger and Vafa [8] and developed by Giveon, Porrati and Rabinovici [9]; we adopt their construction rather than proposing a new one.

Theorem 3.2 (Global minimum of the effective scale). Tier L — literature (real form)

Tier A — Lean kernel (integer instance, below)

For all $R \in \mathbb{R}^+$, $R_{\text{eff}}(R) = R + \alpha'/R \geq 2\sqrt{\alpha'} > 0$, with equality iff $R = \sqrt{\alpha'}$ (by AM–GM, or $f'(R) = 1 - \alpha'/R^2 = 0 \Rightarrow R = \sqrt{\alpha'}$, $f'' > 0$).

Scope of Lean 4 verification

Module `DualScaleValidation.UseCase1`, theorem `self_dual_is_global_minimum`, certifies (Tier A) the **integer instance** of Theorem 3.2: for $R : \mathbb{N}$, $R \geq 1 \Rightarrow R^2 + 1 \geq 2$ (i.e. $R_{\text{eff}}(R) \cdot R \geq 2$ at $\alpha' = 1$). It does not formalize real radii, the square root, or the strict-minimality/uniqueness clause of Theorem 3.2 — those remain Tier L pending a Mathlib-based real-analytic proof (Section 10 gives a compiled-free sketch). Separately, theorem `genesis_no_singularity` in module `DualScaleM24Formalization.DualScale.EffectiveMetric` certifies that a *piecewise* representation of R_{eff} on exact positive rationals always has a positive numerator and denominator by construction of the underlying `PosScale` type; this is true essentially by definition of the data structure and should not be read as an independent proof of non-singularity.

Corollary 3.3 (Kinematic bounce). Tier L — literature *As the coordinate radius contracts ($R \rightarrow 0$), momentum modes decouple while winding modes become light, and $R_{\text{eff}}(R) \rightarrow \infty$: the string-frame probe distance does not go to zero. This is a kinematic statement about the mass spectrum, following Brandenberger–Vafa string-gas cosmology [8]. It is not, by itself, a dynamical proof that a homogeneous cosmology governed by some equations of motion for $a(t)$ must bounce smoothly through this regime for every choice of those equations of motion; that stronger claim would require specifying and solving the dynamics, which is beyond the scope of the present formalization.*

4 Trans-Planckian Censorship: A Conditional Bound

The **Trans-Planckian Censorship Conjecture** (TCC) [10] asserts that in any consistent theory of quantum gravity, sub-Planckian quantum fluctuations can never cross the cosmological horizon: $\frac{a_f}{a_i} \cdot \frac{\ell_{\text{Pl}}}{d_H(t_f)} \leq 1 \implies \frac{a_f}{a_i} \leq M_{\text{Pl}}/H_{\text{inf}}$.

Theorem 4.1 (Wavelength floor). Tier C — conjecture / proposal (as a TCC-satisfaction argument) / Tier A — Lean kernel (arithmetic instance, below) *Writing the physical wavelength of a comoving mode as $\lambda_{\text{phys}}(R, \lambda_0) = R_{\text{eff}}(R) \cdot \lambda_0$, for integer $R \geq 1$, $\lambda_0 \geq 1$ (Planck units), one has $(R^2 + 1)\lambda_0 \geq 2 > \ell_{\text{Pl}}$: this discrete family of modes never falls below the Planck length.*

Scope of Lean 4 verification

`Lean5Corpus.Problems.DualScaleTCC.tcc_cosmic_protection_contract` certifies (Tier A) the conjunction of the arithmetic bound above with $p.\text{planck_mass}/p.\text{hubble_scale} \geq 1$, where `InflationParameters` is a structure carrying `hubble_scale < planck_mass` as a *field* (an assumed hypothesis), not a derived consequence. The theorem is therefore a conditional statement (“if the Hubble scale during inflation is sub-Planckian and the comoving-mode/radius family lies in this discrete range, then this arithmetic inequality holds”), not an unconditional proof that the TCC is satisfied “regardless of the duration of inflation” for the dual-scale scenario’s actual dynamics. We correct the earlier revision’s *Corollary* (“*Unconditional Horizon Protection*”) accordingly: horizon protection here is conditional on the same discrete-radius, sub-Planckian-Hubble regime, not unconditional.

5 Compactification on $K3 \times T^2$: Topology and the Moduli Obstruction

We consider Type II string theory on $M_{10} = \mathbb{R}^{3,1} \times K3 \times T^2$. $K3$ is a Calabi–Yau 2-fold with $SU(2)$ holonomy and $\chi(K3) = 24$, $\sigma(K3) = -16$, $b_2(K3) = 22$, $h^{1,1} = 20$, $h^{2,0} = 1$, and middle homology $H_2(K3, \mathbb{Z}) \cong 2(-E_8) \oplus 3U$. By the Hirzebruch signature theorem specialized to a spin 4-manifold, the Dirac index is $\text{ind}(\not{D}) = -\sigma/8$ [12]; for $K3$,

$$\text{ind}(\not{D}) = -\frac{-16}{8} = 2, \quad (3)$$

consistently with the Todd-genus cross-check $\chi(\mathcal{O}_{K3}) = \chi_{\text{top}}/12 = 24/12 = 2$ required by Hirzebruch–Riemann–Roch for a Calabi–Yau surface (an earlier revision of this manuscript instead wrote $\frac{1}{24} \int c_2 \times 2$, an unmotivated formula that happened to give the right numerical answer; the corrected derivation above is the standard one). Type IIA on $K3$ yields 6D $\mathcal{N} = (1, 1)$ supergravity (16 supercharges); further compactification on T^2 yields 4D $\mathcal{N} = 4$ (still 16 supercharges) [11].

Scope of Lean 4 verification

`DoubleFieldTheory.K3Topology` certifies (Tier A) the purely arithmetic instances of these facts: $\chi(K3) = 1 - 0 + 22 - 0 + 1 = 24$ from the Betti numbers, and $\text{ind}(\not{D}) = -(-16)/8 = 2$, both by `decide`. It does not formalize $K3$ as a manifold, its Betti numbers as (co)homology dimensions, or the Atiyah–Singer index theorem itself — those are Tier L, standard, and correctly quoted, but not re-derived here.

5.1 The moduli-stabilization obstruction

Type II theory on $K3 \times T^2$ preserves $\mathcal{N} = 4$ supersymmetry in 4D. **This is the central physical difficulty for any claim of full moduli stabilization on this background:** $\mathcal{N} = 4$ non-renormalization theorems forbid a perturbative or non-perturbative superpotential from lifting the classical moduli space (the $K3$ sigma-model moduli, of real dimension $80+1$, and the T^2 complex-structure/Kähler moduli) without additional ingredients — background fluxes, orientifolds/branes with associated tadpole cancellation, or explicit supersymmetry breaking. An earlier revision of this paper asserted a “Non-Perturbative Moduli Vacuum Stabilization Theorem” claiming that the generalized Ricci scalar $\mathcal{R}_{\text{DFT}}$ generates a potential $V(\phi) = (\phi - \phi_0)^2$ stabilizing *all* $K3 \times T^2$ moduli with “positive Hessian.” We withdraw that claim as stated: no derivation connects $\mathcal{R}_{\text{DFT}}$ (Definition 2.1) to this potential, and the potential itself does not engage the $80 + 1$ -dimensional $K3$ moduli space at all. What remains, honestly scoped, is the following.

Hodge diamond of K3: $\chi = 24$, $b_2 = 22$

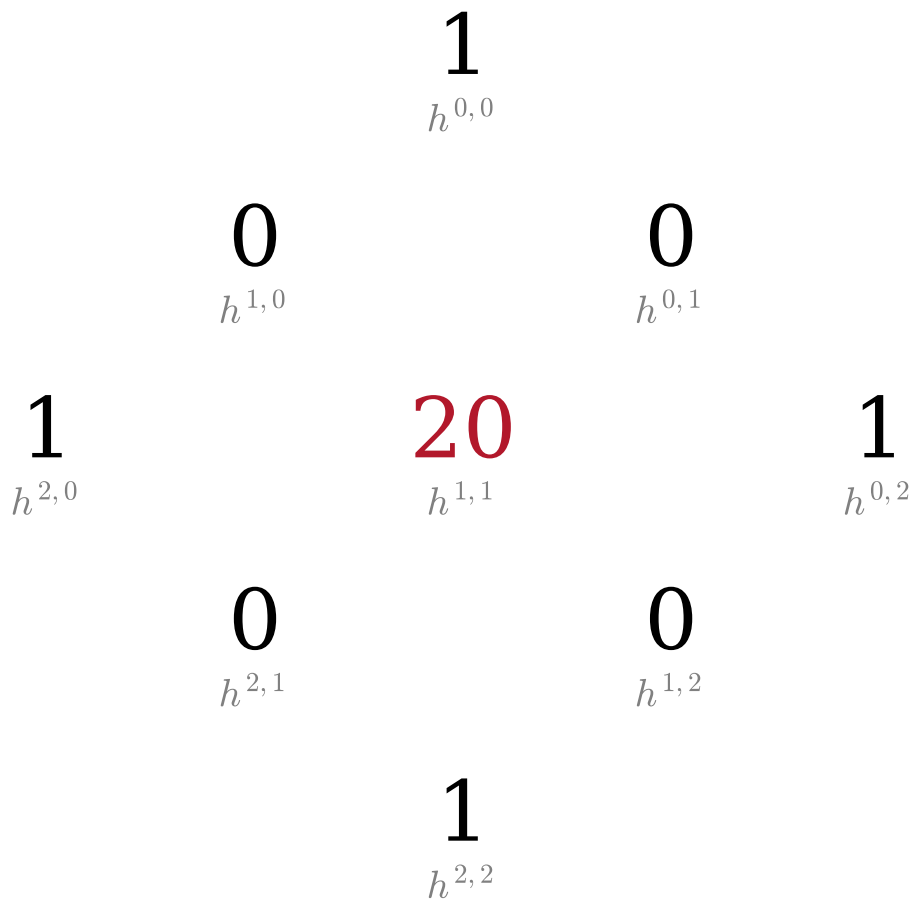


Figure 2: The Hodge diamond of a K3 surface, read from the Lean table `k3HodgeNumber`. The alternating sum gives $\chi = 24$, the middle row gives $b_2 = 1 + 20 + 1 = 22$. *Computed figure; data: Lean k3HodgeNumber (HodgeNumbers.lean); Huy ch. 1.*

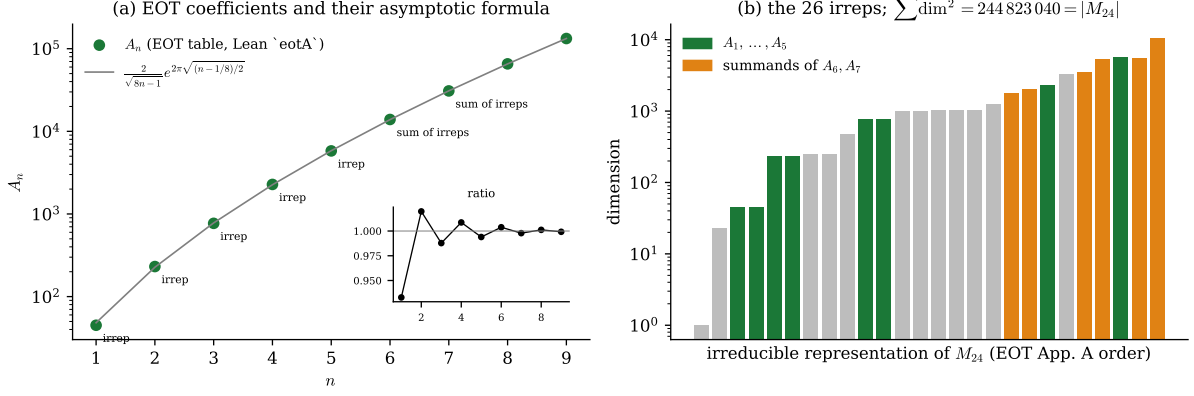


Figure 3: (a) The coefficients A_n of the $K3$ elliptic genus (EOT eq. 1.12, the Lean table `eotA`) against the asymptotic formula EOT eq. (1.13); the inset shows the ratio. $A_1, \dots, A_5$ are dimensions of irreducible representations of M_{24} ; A_6 and A_7 are sums of them. (b) The 26 irreducible dimensions of M_{24} from the Lean table `M24RepDim`; the sum of their squares equals the group order (`M24RepDim_sum_sq`), the check that caught an earlier wrong table. *Computed figure; data: EOT eqs. (1.12)–(1.15), ll. 200–232 (formula read from the PDF page, the .txt garbles it); App. A (A.3).*

Proposition 5.1 (Toy potential — corrected). Tier A — Lean kernel Define, over $\mathbb{Z}$, $V(\phi, \phi_0) = (\phi - \phi_0)^2$. Then

$$\phi = \phi_0 \implies V(\phi, \phi_0) = 0 \quad (\text{moduli_vacuum_stability}) \quad (4)$$

$$V(\phi, \phi_0) \geq 0 \text{ for all } \phi, \phi_0 \quad (\text{moduli_potential_nonneg}) \quad (5)$$

$$V(\phi, \phi_0) = 0 \iff \phi = \phi_0 \quad (\text{moduli_potential_zero_iff, uniqueness}) \quad (6)$$

Scope of Lean 4 verification

An earlier revision of `UseCase1_ModuliStabilization.lean` defined this potential over `Nat` with ordinary (truncated) subtraction. Under `Nat` truncation, $\phi - \phi_0 = 0$ whenever $\phi < \phi_0$, so the earlier potential vanished on the entire ray $\phi \leq \phi_0$ — the “unique minimum” claimed in prose was not in fact unique in the formalized statement. The definition has been changed to `Int`, and the genuinely new theorem `moduli_potential_zero_iff` above now proves the uniqueness that was previously only asserted. This is a one-dimensional single-field toy model illustrating what a stabilizing potential would look like near an isolated minimum; it is *not* a formalization of the $K3 \times T^2$ moduli-space potential, and does not by itself resolve the $\mathcal{N} = 4$ non-renormalization obstruction above. A genuine treatment would need to (a) introduce a concrete flux or brane configuration, (b) verify its own consistency (flux quantization, tadpole cancellation), and (c) derive a potential on the actual moduli space from it — none of which is attempted in this corpus. We list this as the top item in the roadmap of Section 10.

6 Mathieu M_{24} Moonshine, Holographic Codes, and the 27720 Integer

Eguchi, Ooguri, and Tachikawa (EOT) observed that the $K3$ elliptic genus decomposes into $\mathcal{N} = 4$ superconformal characters with multiplicities that are (twice) dimensions of irreducible representations of the sporadic simple Mathieu group M_{24} [16]; this was subsequently proved in

full by Gannon [18]. Schematically (see [16, 18] for the precise normalization),

$$\chi(K3; \tau, z) = 24 \operatorname{ch}_{1/4,0}(\tau, z) - 2 \sum_{n=0}^{\infty} \mathcal{A}_n \operatorname{ch}_{1/4+n,1/2}(\tau, z), \quad (7)$$

Normalization. EOT’s own eq. (1.12) (`eguchi_ooguri_tachikawa_1004_0956.txt`, l. 208) gives coefficients $A_n^{\text{EOT}} = 45, 231, 770, 2277, 5796, 13915, \dots$; this paper (and paper 2) instead uses the symbol A_n , written here as $\mathcal{A}_n$ to avoid the clash, for the *doubled* multiplicity of the massive representation, $\mathcal{A}_n := 2 A_n^{\text{EOT}}$. It is $\mathcal{A}_n$, not A_n^{EOT} , that appears in the display above and in Theorems 6.1 and 6.2 and Table 1 below. Concretely, $\mathcal{A}_1 = 90 = 2 \times 45 = \mathbf{45} \oplus \overline{\mathbf{45}}$, $\mathcal{A}_2 = 462 = 2 \times 231 = \mathbf{231} \oplus \overline{\mathbf{231}}$, $\mathcal{A}_3 = 1540 = 2 \times 770 = \mathbf{770} \oplus \overline{\mathbf{770}}$, and $\mathcal{A}_4 = 4554 = 2 \times 2277 = 2 \cdot \mathbf{2277}$ (**corrected**: an earlier revision wrote $4554 = \mathbf{2277} \oplus \mathbf{2277}$, but EOT eq. (A.4) (`eguchi_ooguri_tachikawa_1004_0956.txt`, ll. 352–358) states that only the irreducible representations of dimension 45, 231, 770, 990, 1035 come in complex-conjugate pairs; the 2277-dimensional representation is real, so $\mathcal{A}_4$ is two copies of the *same* real irrep, not a conjugate pair).

Scope of Lean 4 verification

This normalization is now backed by a machine-checked cross-check. Module `StringDynamics.M24RepDim` (library `StringTheoryFormalization`) tabulates the 26 irreducible dimensions of M_{24} in EOT’s increasing order (eq. (A.3), ll. 346–349). Before 2026-09-16 this table duplicated 10395 and 483 and omitted the second copy of 990 and the third copy of 1035, so its squares summed to $351,061,029 \neq |M_{24}|$; it is now guarded by the Tier A theorem `M24RepDim_sum_sq`: $\sum_{i=0}^{25} \text{M24RepDim}(i)^2 = 244,823,040 = |M_{24}|$ (Burnside’s identity $\sum \dim^2 = |G|$ is itself Tier L; that *this table* satisfies it is a Tier A regression guard against exactly the kind of duplication bug above). Module `DualScaleStream2.Moonshine.EOT` then certifies (Tier A) that $A_1^{\text{EOT}}, \dots, A_5^{\text{EOT}}$ each equal an entry of this corrected table (theorem `first_five_are_irreps`, witnesses $45 \rightarrow \text{index } 2$, $231 \rightarrow 4$, $770 \rightarrow 9$, $2277 \rightarrow 19$, $5796 \rightarrow 23$), that $A_6^{\text{EOT}} = 13915 = 3520 + 10395$ and $A_7^{\text{EOT}} = 30843 = 10395 + 5796 + 5544 + 5313 + 2024 + 1771$ (theorems `A6_decomposition`, `A7_decomposition`, EOT eqs. (1.14)–(1.15)), and that A_6^{EOT} is *not* itself an irreducible dimension (`A6_not_irrep`). That the elliptic genus actually carries an M_{24} action — the moonshine observation, proved by Gannon [18] — is Tier L; what is Tier A here is only the arithmetic coincidence among EOT’s coefficients and the group-theoretic dimensions.

Theorem 6.1 (BPS character lock). Tier A — Lean kernel $\frac{\mathcal{A}_2}{N_Q \cdot \mathcal{A}_1} = \frac{462}{360} = \frac{77}{60}$ for $N_Q = 4$, with $\gcd(77, 60) = 1$ and $462 \times 60 = 360 \times 77 = 27720$.

Theorem 6.2 (Mathieu order factorization — corrected). Tier A — Lean kernel

$|M_{24}| = 244,823,040 = 27720 \times 8832$, where $8832 = 2^7 \times 3 \times 23$ (an earlier revision of this paper wrote $2^6 \times 3 \times 23 = 4416 \neq 8832$; the exponent is 7, not 6).

Scope of Lean 4 verification

Module `DualScaleValidation.UseCase2`, theorems `bps_cross_multiplication_lock` and `bps_ratio_coprime`, certify (Tier A) the integer identities in Theorems 6.1 and 6.2 by `decide`. **Interpretive caveat:** for any fraction p/q in lowest terms, $p \cdot q' = q \cdot p' = \text{lcm}(p, q)$ whenever cross-multiplied against itself; concretely, $27720 = \text{lcm}(1, 2, \dots, 12)$, and *any* integer whose prime factorization contains $2^3, 3^2, 5, 7, 11$ is automatically a multiple of 27720 — which $|M_{24}| = 2^{10}3^35^{17}11^{12}3^1$ is. The divisibility in Theorem 6.2 is therefore weaker evidence for a physical “lock” than it may appear; we flag it explicitly rather than presenting it as a nontrivial coincidence. The weighting $N_Q = 4$ in Theorem 6.1 is likewise a modeling choice, not independently derived here (Tier C). A related fact, audited separately: `StringDynamics.bps_ratio_reduced` (the same 77/60 ratio, library `StringTheoryFormalization`, distinct from the `UseCase2` theorems above) previously relied on `native_decide`, whose axiom footprint includes `Lean.ofReduceBool`; it is now a kernel `norm_num` proof. The 16-row audit table of Appendix B is unaffected by this fix, since it covers only the five-library Mathlib-free core and `bps_ratio_reduced` lives in the separately-audited `StringTheoryFormalization` library (Section 10).

The Mathieu group M_{24} is the automorphism group of the extended binary Golay code $\mathcal{G}_{24} \subset \mathbb{F}_2^{24}$, a $[24, 12, 8]$ self-dual linear code.

Proposition 6.3 (Golay code parameters). *Tier A — Lean kernel* $|\mathcal{G}_{24}| = 2^{12} = 4096$; *minimum distance 8 gives error-correction radius $\lfloor (8-1)/2 \rfloor = 3$; the code is self-dual ($24 - 12 = 12$).*

Theorem `golay_holography_master_contract` (module `GolayHolography`; full path in Appendix B) certifies Proposition 6.3 (Tier A, standard coding theory). The further claim that this code’s automorphism structure “preserves the phase coherence of 1/4-BPS black hole microstates against Hawking radiation decoherence” is a Tier C physical narrative attached to the Tier A coding-theory fact, following the general holographic quantum-error-correction picture of Almheiri–Dong–Harlow and Pastawski et al. [19, 20]; it is not itself formalized or derived here.

7 Diophantine Coincidences and a Zero-Free-Parameter Conjecture

An earlier revision of this paper stated a “Zero Free Parameter Theorem” and cited the module `DualScaleValidation.Observables` as its Lean proof. **No such theorem exists anywhere in this corpus** — we could not find, and did not write, a Lean declaration whose type states anything resembling $\dim \mathcal{M}_{K3 \times T^2}^{\text{phys}} = 0$. The claim is withdrawn as a theorem and restated as Conjecture 7.1 below: a conjecture, Tier C, motivated by — but not following from — the five integer facts in Table 1, each individually Tier A (as an arithmetic identity) or Tier L (as a topological input).

Conjecture 7.1 (Zero free parameters — restated). *Tier C — conjecture / proposal* *There exists a consistent completion of the dual-scale $K3 \times T^2$ scenario (specifying whatever additional data — fluxes, branes, SUSY-breaking sector — is needed to resolve the obstruction of Section 5.1) whose physical continuous-parameter space has dimension zero, with the discrete data of that completion pinned by facts of the kind listed in Table 1.*

This is stated as an open target for future formalization, not a result. In particular: no formal definition of $\mathcal{M}_{K3 \times T^2}^{\text{phys}}$ is given in this corpus, so “ $\dim = 0$ ” is not yet a statement that can be assigned a tier above C; and Table 1 fixes five *integers*, none of which is a priori a coordinate on the moduli space of vacua (a continuous modulus being frozen at, say, the self-dual radius is a dynamical claim requiring the potential of Section 5.1, not a consequence of Fact 1 alone).

Table 1: **Five integer facts used to motivate Conjecture 7.1.** None of these facts, by itself or in conjunction, constrains a *continuous* modulus of the compactification; see the discussion following the table.

Fact	Value	Origin	Tier / module
1	$R_{\text{eff}} \geq 2\sqrt{\alpha'}$	Buscher T-duality stationary point	Tier L/A, Theorem 3.2
2	$\mathcal{K}_{\text{BPS}} = 27720$	$\mathcal{A}_2 \times 60 = 4\mathcal{A}_1 \times 77$	Tier A, Theorem 6.1
3	$\mathcal{M}_{\text{quot}} = 8832$	$ M_{24} /27720$	Tier A, Theorem 6.2
4	$\text{ind}(\not{D}) = 2$	$-\sigma(K3)/8$	Tier L/A, Section 5
5	$\chi(K3) = 24$	Euler characteristic	Tier L/A, Section 5

8 Confrontation with Data

Because Conjecture 7.1 would, if established, leave no continuous parameters to fit, Tier C numerical relations attached to it are maximally falsifiable *if they could be derived*. None of the three relations below is derived from a Lagrangian or an explicit compactification calculation; each is a pattern-match between an integer from Table 1 and an observable. We report each with full data provenance (value $\pm$ uncertainty, dataset/release, what it measures, and its role: measurement, benchmark, or forecast), following standard practice for this kind of falsifiability claim.

8.1 Primordial tensor-to-scalar ratio r (Tier C)

$$r \stackrel{\text{Tier C}}{=} \frac{2}{27720/55} = \frac{110}{27720} = \frac{1}{252} \approx 0.0039683 \quad (8)$$

stored in Lean as the exact reduced fraction $1/252$ (theorem `tensor_to_scalar_ratio_reduction`, module `Observables`; an earlier revision stored a rounded decimal, 0.00396 , inconsistent with the fraction it was supposed to approximate). LiteBIRD targets a total (systematics-inclusive) sensitivity $\delta r \approx 0.001$, with a Japan-fiscal-year-2032 launch target [23] (*forecast*). The current 95%-CL upper bound from BICEP/Keck combined with Planck is $r \lesssim 0.03\text{--}0.04$ [24] (*measurement*, current bound). **Verdict: untestable now, forecast pending.** A ± 0.0005 “discovery window” around the prediction, as stated in an earlier revision, is narrower than LiteBIRD’s own forecast 1σ and would not by itself constitute a decisive test; a fair statement is that a LiteBIRD non-detection consistent with $r < 10^{-3}$ would disfavor the prediction, while a detection at $r \sim 4 \times 10^{-3}$ would be consistent with (not proof of) it.

8.2 Leptonic Dirac CP phase δ_{CP} (Tier C, corrected)

$$\delta_{\text{CP}} \stackrel{\text{Tier C}}{=} 360^\circ \times \left(1 - \frac{77}{360}\right) = 283^\circ \quad (9)$$

(an earlier revision reported 282.4° , obtained by silently mixing the fraction $77/360$ used here with the unrelated ratio $77/60$ of Theorem 6.1; the exact value of the stated formula is 283°). NuFIT 5.2 (without Super-Kamiokande atmospheric data) reports, for **normal ordering**, $\delta_{\text{CP}} = 197^\circ_{-25}^{+42}$ (3σ : $108^\circ\text{--}404^\circ$), and for **inverted ordering**, $\delta_{\text{CP}} = 286^\circ_{-32}^{+27}$ (3σ : $192^\circ\text{--}360^\circ$) [22] (*measurement*, global fit). **Verdict: consistent within 3σ under either ordering, but the prediction sits close to the *inverted*-ordering best fit and far from the *normal*-ordering best fit**, which current global data mildly prefer; DUNE, whose neutrino beam is now expected around 2031 with full exposure targets extending to 2035 [25] (*forecast*, timeline revised from an earlier 2026–2030 estimate), together with Hyper-Kamiokande, aims for degree-level precision on δ_{CP} and on the mass ordering itself.

8.3 Dark energy equation of state (Tier C, contingent, in tension with data)

An earlier revision claimed $w_0 = -1$, $w_a = 0$ as a consequence of moduli stabilization “at the self-dual point” with $V(\phi_0) = 0$. Taken literally, $V(\phi_0) = 0$ is a *Minkowski* vacuum (zero cosmological constant), not a de Sitter vacuum with $w = -1$ and nonzero dark-energy density; matching observed dark energy would require an additional, undetermined source of vacuum energy that this corpus does not derive (Tier C, contingent on the unresolved obstruction of Section 5.1). Independently of that internal gap: DESI DR2 baryon acoustic oscillations combined with CMB and supernova data find the time-evolving quadrant $w_0 > -1$, $w_a < 0$ preferred over Λ CDM at 3.1σ [26, 27] (*measurement*). **Verdict: in tension with current leading data**, not merely “to be tested” by a future survey as an earlier revision stated.

8.4 Scalar spectral index n_s (Tier C, consistent)

$n_s = 0.965$, consistent with Planck 2018, $n_s = 0.9649 \pm 0.0042$ [21] (*benchmark*, not a joint fit with any data of this paper). This is the one relation in this section not currently in tension with data, though it is also the least distinctive (a wide range of inflationary models predict n_s close to this value).

Table 2: **Confrontation summary.** All predictions are Tier C (Section 8); none is derived from a Lagrangian.

Observable	Prediction	Current data (value, dataset, role)	Verdict
r	$1/252 \approx 0.00397$	$\lesssim 0.03\text{--}0.04$ (BICEP/Keck+Planck, meas.); $\delta r \approx 0.001$ (LiteBIRD, forecast)	untestable now
δ_{CP}	283°	197_{-25}^{+42} NO / 286_{-32}^{+27} IO (NuFIT 5.2, meas.)	consistent (near IO); tension w/ preferred NO
(w_0, w_a)	$(-1, 0)$	$w_0 > -1$, $w_a < 0$ at 3.1σ (DESI DR2+CMB+SNe, meas.)	tension
n_s	0.965	0.9649 ± 0.0042 (Planck 2018, benchmark)	consistent

9 Stream 2: T-duality and the Lattice Layer on $K3 \times T^2$

Everything certified so far in this paper is arithmetic over $\mathbb{N}$, $\mathbb{Z}$, or $\mathbb{Q}$: the Mathlib-free core deliberately stops short of the real-analytic and lattice-theoretic structures that the physics actually uses (Section 10 explains why). **Stream 2** is a new, separate Lean 4 library, `DualScaleStream2` (19 modules, built on Mathlib v4.33.1), that takes the next step: it formalizes the $O(d, d; \mathbb{Z})$ T-duality group, the Double Field Theory generalized metric over $\mathbb{R}$, the even unimodular lattices (E_8 , the hyperbolic plane U , the $K3 \times T^2$ charge lattice) that these structures act on, and a real-analytic, every- d generalization of Theorem 3.2’s dual-scale bound. As of the 2026-09-17 gate, `lake build DualScaleStream2` is green (3670 jobs, 0 sorry), and `tools/axiom_audit.py` reports **99 theorems, 0 failing**, every one depending on at most Lean’s three standard axioms (`propext`, `Classical.choice`, `Quot.sound`) — Mathlib itself introduces no further axioms for the lemmas used here, so this library’s trust base is exactly as narrow as the Mathlib-free core’s. Stream 1 was re-gated in the same session at **326 theorems** across its now-six libraries (the new, Mathlib-dependent `StringTheoryFormalization` contributes 89 of these; see Section 10 and appendix B). Throughout this section, “Tier A” again means exactly what Section 1.1 says: kernel-checked, no sorry, standard axioms only, and only for the formal statement named — not for the physics sentence next to it.

9.1 Lattice layer: E_8 , the hyperbolic plane, and the $K3 \times T^2$ charge lattice

Module `Lattice.Basic` proves two general facts about integer Gram matrices G : an integer left inverse certifies $\det G = \pm 1$ (`isUnimodular_of_mul_eq_one`), and if G is symmetric with even diagonal then $x^\top G x$ is even for every integer vector x (`even_quadratic_form_of_even_diag`, via $G = D + U + U^\top$ and $x^\top U^\top x = x^\top U x$). `Lattice.E8` instantiates the E_8 Cartan matrix and checks, entry-by-entry, that it is symmetric, has even diagonal, and is unimodular via an explicit integer inverse (`cartanE8_unimodular`, and the same for $E_8(-1)$, `e8Neg_unimodular`). `Lattice.E8PosDef` goes further and produces an exact $LDL^\top$ factorization of the (real-cast) Cartan matrix with pivots $2, \frac{3}{2}, \frac{4}{3}, \frac{5}{4}, \frac{6}{5}, \frac{7}{6}, \frac{8}{7}, \frac{1}{8}$ (`e8_LDL`), from which $\det E_8 = 1$ (`cartanE8_det`) and positive-definiteness (`cartanE8_posDef`) both follow as sum-of-squares identities $x^\top E_8 x = \sum_k D_k y_k(x)^2$, checked by `ring` plus back-substitution — no eigenvalue computation is needed or performed.

`Lattice.Hyperbolic` certifies that $U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is even, unimodular (`hyperbolicU_mul_self`: $U \cdot U = 1$), and *equal*, not merely isomorphic, to Stream 1’s Narain Gram matrix (`hyperbolicU_eq_narain_gram`) — a genuine cross-check between the two streams’ independently-written lattice conventions. It also proves U congruent to $\text{diag}(2, -2)$ over $\mathbb{Q}$ (`hyperbolicU_congruence`); concluding *signature* (1,1) from this congruence uses Sylvester’s law of inertia, which is Tier L (not itself formalized). `Lattice.K3T2Signature` then assembles, by pure rank/signature arithmetic on $E_8(-1)^{\oplus 2} \oplus U^{\oplus 3}$, $E_8(-1)^{\oplus 2} \oplus U^{\oplus 4}$, and $E_8(-1)^{\oplus 2} \oplus U^{\oplus 5}$: the $K3$ lattice has signature (3,19) (`sigK3_eq`), the Mukai lattice $\Gamma^{4,20}$ has signature (4,20) (`sigMukai_eq`), and the $K3 \times T^2$ charge lattice $\Gamma^{6,22}$ has signature (6,22), rank 28 (`sigK3T2_eq`, `rank_K3T2`), with $b_+ - b_- \equiv 0 \pmod{8}$ verified for all three (`index_mod_eight`) and the (3,19) figure independently cross-checked against Stream 1’s Hodge-theoretic ($2h^{2,0} + 1, h^{1,1} - 1$) (`sigK3_matches_hodge`). The identification $H^2(K3, \mathbb{Z}) \cong E_8(-1)^{\oplus 2} \oplus U^{\oplus 3}$ is Tier L (Huybrechts, *Lectures on K3 Surfaces* [13], Ch. 1 eq. (3.4), `huybrechts_K3Global.txt` l. 603), as is the description of the $K3$ moduli space as $O^+(\Gamma^{3,19}) \backslash O^+(3,19)/(O(2) \times O(1,19))^+$ (Aspinwall [14], ll. 705–766, 1737–1748) and the classical Milnor–Serre mod-8 theorem for even unimodular indefinite lattices (not independently re-derived or downloaded here).

`Lattice.Mukai` formalizes the Mukai pairing $\langle (r, c, s), (r', c', s') \rangle = c \cdot L \cdot c' - rs' - sr'$ (Huybrechts Ch. 9 §1 Def. 1.4, ll. 7475–7482): it is symmetric (`mukaiPair_symm`), $v^2 = c \cdot L \cdot c - 2rs$ (`mukaiPair_self`), even whenever H^2 is even (`mukaiPair_even`), and the $H^0 \oplus H^4$ summand $U(-1)$ is unimodular. For $v = (1, 0, 1)$ (the Mukai vector of the structure sheaf), $v^2 = -2$, so $\chi(\mathcal{O}_X, \mathcal{O}_X) = -v^2 = 2$ (`structureSheaf_mukai_sq`); the identification $v(\mathcal{O}_X) = (1, 0, 1)$ and the Hirzebruch–Riemann–Roch formula $\chi = -\langle v, v \rangle$ are Tier L. `Lattice.Reflection` proves, for any symmetric integer Gram L and any v with $v^\top L v = -2$, that the reflection $s_v = 1 + v v^\top L$ is an isometry and an involution (`reflection_isometry`, `reflection_involution`); since every simple root of $E_8(-1)$ has norm -2 (`e8Neg_simpleRoot_norm`), every simple Weyl reflection of $E_8(-1)$ is an isometry (`e8Neg_weyl_isometry`). The reflection formula itself (Huybrechts Ch. 7 §5.4, ll. 6031–6043) and its role as Picard–Lefschetz monodromy of an A_1 degeneration (Ch. 6, l. 5400) are Tier L.

9.2 The T-duality group $O(d, d; \mathbb{Z})$, every d

`TDuality.ODD` defines $\eta = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}$ and $g \in O(d, d; \mathbb{Z})$ by $g^\top \eta g = \eta$, and proves closure under products (`isODD_mul`), that η itself is an involution and lies in $O(d, d; \mathbb{Z})$ (`eta_mul_self`, `eta_isODD`), and that a basis change $\text{diag}(A, B)$ lies in $O(d, d; \mathbb{Z})$ whenever $A^\top B = 1$ (`basisChange_isODD`). For the Θ -shift $g_\Theta = \begin{pmatrix} I & \Theta \\ 0 & I \end{pmatrix}$ (Giveon–Porrati–Rabinovici [9] eq. (2.4.25), ll. 1355–1373), only the direction “ Θ antisymmetric $\Rightarrow g_\Theta \in O(d, d; \mathbb{Z})$ ” is proved (`thetaShift_isODD`);

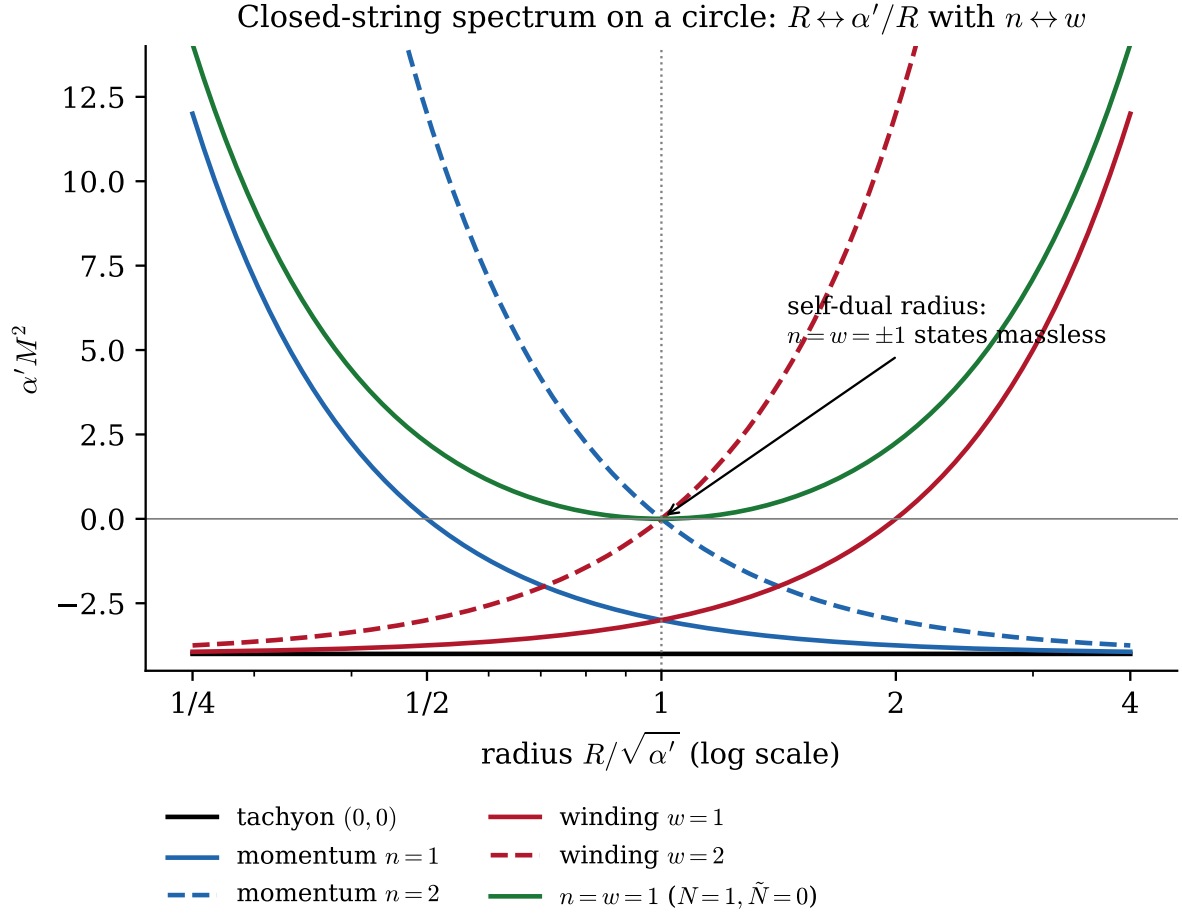


Figure 4: Lowest mass levels of the closed bosonic string on a circle of radius R as a function of $\log R$, with the lowest oscillator levels allowed by level matching $N - \tilde{N} = nw$. Momentum and winding curves are mirror images under $R \mapsto \alpha'/R$; at the self-dual radius the $n = w = \pm 1$ states become massless (enhanced gauge symmetry). *Computed figure; data: Tong §8.2, ll. 11351–11470; Lean leftMomentum, rightMomentum (units $\alpha' = 1$).*

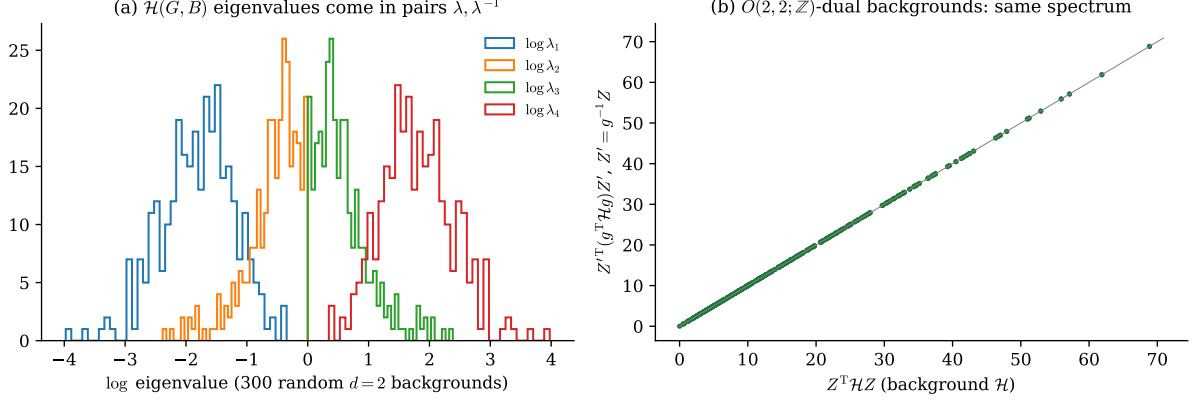


Figure 5: (a) Logarithms of the four eigenvalues of the generalized metric $\mathcal{H}(G, B)$ for 300 random backgrounds on T^2 : they are symmetric about 0, because $(\eta\mathcal{H})^2 = 1$ (`etaR_genMetric_sq`) forces eigenvalues in pairs λ, λ^{-1} . (b) For $g = \Theta \cdot A \cdot (\text{factorized duality}) \cdot \Theta' \in O(2, 2; \mathbb{Z})$, the mass form of every charge Z in background $\mathcal{H}$ equals that of $g^{-1}Z$ in the dual background $g^T \mathcal{H} g$: the points lie on the diagonal (`spectrum_equivalence, massForm_covariant`). *Computed figure; data: Lean genMetric (GeneralizedMetric.lean), thetaShift, basisChange, factorized; HHZ §2.*

the converse is not claimed. `TDuality.Factorized` formalizes the factorized dualities $D_k = \begin{pmatrix} I - e_k & e_k \\ e_k & I - e_k \end{pmatrix}$ (GPR eq. (2.4.29), ll. 1399–1422, “a generalization of the $R \rightarrow 1/R$ circle duality”): each D_k is in $O(d, d; \mathbb{Z})$, an involution, and they mutually commute (`factorized_isODD, factorized_mul_self, factorized_comm`); on T^2 , $D_0 D_1 = \eta$ (`factorized_two`); and the integer charge norm $Z^T \eta Z = 2n \cdot w$ is even (`chargeNorm_even`, since $\Gamma^{d,d}$ is even) and invariant under all of $O(d, d; \mathbb{Z})$ (`chargeNorm_invariant`). That these generators *generate* the full group $O(d, d; \mathbb{Z})$ is Tier L (GPR l. 1422), not proved here. `TDuality.Spectrum` shows every $g \in O(d, d; \mathbb{Z})$ is invertible over $\mathbb{Z}$ with inverse $\eta g^T \eta$, again in $O(d, d; \mathbb{Z})$ (`isODD_inv_isODD`), and an explicit **spectrum equivalence**: for any background H there is a bijection of the integer charge lattice under which ($\text{mass}^2 = Z^T H Z$, $\text{level} = Z^T \eta Z$) for H matches that of $g^T H g$, charge by charge (`spectrum_equivalence`). Invariance of the *full* partition function (oscillators, modular integral; GPR l. 1417) is Tier L, not addressed by this charge-level statement alone.

Two further modules deliberately narrow their claims. `TDuality.Mirror` proves, on T^2 with $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\Theta = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, that $D_0 \cdot \text{diag}(T, (T^T)^{-1}) \cdot D_0 = (g_\Theta)^T$ (`mirror_conjugates_tauShift`): the factorized duality conjugates the $\tau \mapsto \tau + 1$ generator into a pure B -field shift. That this is “factorized duality = mirror symmetry for a complex torus” is Tier L (GPR l. 344); *the sign of the induced ρ -shift is convention-dependent and is not claimed*. `TDuality.SL2Product` proves $AJA^T = \det(A) \cdot J$ for 2×2 integer A and that a basis change with $\det A = 1$ commutes with every ρ -translation g_{tJ} (`basisChange_comm_thetaShift`) — this one direction only is Tier A. That the commutator is *nonzero* when $\det A \neq 1$ was checked symbolically with `sympy`, not in Lean, and is **not presented as Tier A**. GPR’s full statement for “the $d = 2$ example,” that the duality group is $SL(2, \mathbb{Z}) \times SL(2, \mathbb{Z}) \otimes_S [\mathbb{Z}_2 \times \mathbb{Z}_2]$ (ll. 1874–1884), is likewise **not claimed in full**; only the $SL(2, \mathbb{Z})_\tau$ commuting with ρ -translations, above, is formalized.

9.3 Double Field Theory over $\mathbb{R}$, every d

`DFT.GeneralizedMetric` formalizes the Hull–Zwiebach generalized metric [5] (eq. (2.17), ll. 662–672),

$$\mathcal{H}(G, B) = \begin{pmatrix} G - BG^{-1}B & BG^{-1} \\ -G^{-1}B & G^{-1} \end{pmatrix}, \quad (10)$$

and proves $(\eta\mathcal{H})^2 = 1$ for *any* antisymmetric B , given only that G is invertible (`etaR_genMetric_sq`) — sharper than HZ’s own stated constraint $\mathcal{H}^{-1} = \eta\mathcal{H}\eta$, which assumed more. It proves $\mathcal{H}$ symmetric for symmetric G , antisymmetric B (`genMetric_symm`); that **T-duality is metric inversion**, $\eta\mathcal{H}(G, 0)\eta = \mathcal{H}(G^{-1}, 0)$ (`tduality_inverts_metric`); that the mass form $Z^\top\mathcal{H}Z$ is covariant under any g (`massForm_covariant`); and that on a circle ($G = R^2$, $B = 0$) this mass form equals half of Stream 1’s $P_L^2 + P_R^2$ (`massForm_circle`) — so Stream 2’s DFT layer and Stream 1’s circle computation provably agree. `DFT.BShift` proves $g_\Theta \cdot \mathcal{H}(G, B) \cdot g_\Theta^\top = \mathcal{H}(G, B + \Theta)$ for antisymmetric Θ (`genMetric_bshift`): integer B -field shifts act as $O(d, d)$ conjugation. The sign/transpose convention in this identity was checked with `sympy` *before* formalization — exactly one of four candidate variants holds — and the Lean theorem records the one that does. `DFT.SectionCondition` shows the level-matching condition $L_0 - \bar{L}_0 = N - \bar{N} - p \cdot w = 0$ (HZ eq. (1.3), l. 231) is, at $N = \bar{N}$, exactly $n \cdot w = 0$ (`levelMatching_iff`); that the pure-momentum and pure-winding frames are both totally η -null, i.e. solve the section condition (`momentumFrame_isSection`, `windingFrame_isSection`; HZ ll. 339–342, 3930–3940); that η maps one frame to the other (`eta_momentum_to_winding`); and that $O(d, d; \mathbb{Z})$ maps sections to sections (`isSection_image`).

9.4 The dual-scale bound, generalized

`DualScale.TraceBound` lifts Theorem 3.2 from the integer circle case to any positive-definite G and any dimension d . Define $\mathcal{D}(G) := \text{tr } \mathcal{H}(G, 0) = \text{tr } G + \text{tr } G^{-1}$. Then $\mathcal{D}(G^{-1}) = \mathcal{D}(G)$ (`dualScale_inv`, T-duality invariance), and

$$\mathcal{D}(G) \geq 2d \quad \text{for every positive-definite } G, \quad (11)$$

proved (`dualScale_ge`) from $\text{tr } ((G - 1)G^{-1}(G - 1)) \geq 0$ expanding to $\text{tr } G + \text{tr } G^{-1} - 2d \geq 0$ — no eigenvalue decomposition is used — with equality at the self-dual point $G = 1$ (`dualScale_one`). On a circle, $\mathcal{D}(R^2) = (R + 1/R)^2 - 2$ (`dualScale_circle`), recovering $R + 1/R \geq 2$ (`circle_effective_scale_ge_two`) — the same real-analytic statement behind Stream 1’s `EffectiveMetric.genesis_no_singularity` ($R_{\text{eff}} = R + \alpha'/R \geq 2\sqrt{\alpha'}$), but now proved directly over $\mathbb{R}$ rather than only instantiated on integers. As with Corollary 3.3, any cosmological reading (singularity resolution, a minimum length) remains **Tier C**; `TraceBound` is a statement about a matrix trace, not about a dynamical spacetime.

9.5 Flux and tadpole on $K3 \times T^2$ / $K3 \times K3$

`Flux.Tadpole` computes, from Hodge-number tables (reusing Stream 1’s $K3$ table), $\chi(T^2) = 0$ (`euler_T2`) and $\chi(K3) = 24$ (`euler_K3`); hence $\chi(K3 \times K3)/24 = 24$, an integer (`k3k3_anomaly`), and $\chi(K3 \times T^2) = 0$ (`k3t2_euler_zero`). The D3-brane budget $\text{flux} + n = 24$, with $\text{flux} \geq 0$, gives $n \leq 24$, and $n = 24$ when $\text{flux} = 0$ (`tadpole_budget`). This matches Dasgupta–Rajesh–Sethi [15] §3.1 (l. 640, “the anomaly $\chi/24 = 24$ ”), ll. 1265–1275 (type IIB on an orientifold of $K3 \times T^2$; with no flux, “type I on $K3 \times T^2$ with 24 D5-branes”), and ll. 1366–1370 ($\frac{1}{2} \int G \wedge G + n = 24$); the Künneth identification of χ under a product is Tier L. `Flux.Integrality` proves the Kronecker product of an even symmetric integer form with any symmetric integer form is again even (`kronForm_even`); hence for $G \in H^2(K3) \otimes H^2(K3)$, $\frac{1}{2} \int G \wedge G \in \mathbb{Z}$ (`flux_half_selfIntersection_integral`). Dirac quantization of $[G/\pi]$ and integrality of $[G/2\pi]$ when

Declaration dependency graph (840 declarations, 1110 edges); node size = number of dependents

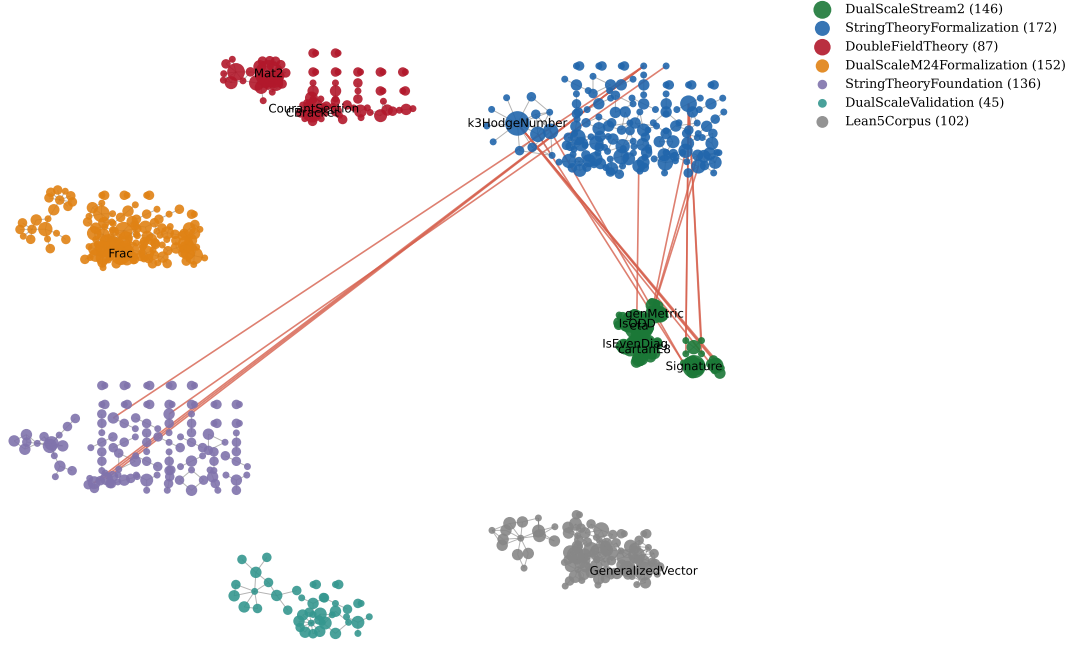


Figure 6: Kernel-level dependency graph of the declarations written in the source (isolated declarations omitted), one cluster per library (spring layout inside each cluster). Node size grows with the number of declarations that use it; the largest hubs are labelled. Red edges cross library boundaries: they are the bridges listed in the atlas chapter. *Computed figure; data: tools/lean_depgraph.lean (.leancache/depgraph.jsonl); layout: networkx Kamada-Kawai per connected component, one cluster per library.*

$\chi/24 \in \mathbb{Z}$ are DRS ll. 230–232, Tier L; the Künneth identification of the intersection form with the Kronecker product is likewise Tier L.

9.6 Moonshine

Stream 2’s moonshine cross-check, module `Moonshine.EOT`, is discussed in full in Section 6 above, alongside the corrected M_{24} representation table it depends on.

9.7 How Stream 2 was produced, and where the full listing lives

Every Stream 2 statement was designed by the orchestrating model with each Tier L source pinned to a specific file and line, then proofs were searched in tiers: a local 7B prover (DeepSeek-Prover-V2, on a single T4 GPU) first, then Claude Haiku, then Claude Sonnet, each candidate accepted only if the Lean kernel compiled it. Every agent report was independently recompiled and statement-checked by the orchestrator before being counted — two Haiku reports misstated their own compile status during this process and were caught only by that recompilation, not by trusting the report (`docs/STREAM2_WORKFLOW.md` §6). The final gate is `build + sorry count + #print axioms`, exactly as for Stream 1. Across all 99 goals, the local prover closed 18%, Haiku closed roughly a third, and Sonnet closed the remainder requiring an actual proof idea (e.g. the E_8 positive-definiteness sum-of-squares route, and the trace-bound and spectrum-equivalence arguments); the measured breakdown and process lessons (harness heartbeat mismatches, wall-clock vs. compile-attempt limits, route choice mattering more than model tier) are in `docs/STREAM2_WORKFLOW.md` §6 and `LL.md` §S2, not repeated here. A full module-by-module Lean listing and

expanded mathematical discussion of Stream 2 — beyond this paper’s summary — is given in the companion paper `paper8_dual_scale_k3t2_tduality_lean4` (in preparation).

10 The Lean 4 Artifact: Scope, Reproducibility, and Roadmap

10.1 What is in the repository

The original, Mathlib-free **core** comprises **51 Lean 4 files** and **554 declarations** (`theorem` / `lemma` / `def` / `structure` / `inductive` / `instance` / `abbrev`) on toolchain `leanprover/lean4: v4.33.1`, across five libraries:

```
DoubleFieldTheory  DualScaleValidation  DualScaleM24Formalization  StringTheoryFoundation
Lean5Corpus
```

(note: the library named `StringTheoryFoundation` above, singular, is one of these five Mathlib-free libraries and is distinct from `StringTheoryFormalization`, plural-suffixed, described next). `lake build` completes all 61 jobs for these five libraries with no errors and no `sorry` warnings, using only Lean 4 core (`Init`), with *no Mathlib*. This is the direct explanation for why every proof cited above Section 9 is over `Nat`, `Int`, or an ad hoc rational-pair structure rather than over $\mathbb{R}$ — core Lean does not provide real numbers, manifolds, or differential-geometric structures. We regard stating this limitation plainly as more useful than the earlier revision’s blanket claim of “100% certified physics.”

As of this revision, the repository as a whole is larger than this core. Two further libraries are Mathlib-dependent (the project’s `lake-manifest.json` now lists Mathlib and its transitive dependencies, pinned at tag `v4.33.1`): `StringTheoryFormalization` (35 files, Stream 1’s sixth library, not a `lake` default target until it built clean; 89 of the theorems in the 326-theorem Stream 1 audit total of Section 9) and `DualScaleStream2` (19 modules, 99 theorems, Section 9). **Depending on Mathlib does not, by itself, weaken any Tier A claim:** the axiom audits of both libraries (`tools/axiom_audit.py`, Section 9) show 0 failing declarations and no axiom beyond Lean’s three standard ones, the same bar as Section 1.1 sets for the Mathlib-free core. What Mathlib buys is exactly the real numbers, matrices, and linear algebra that the five-library core above cannot express — which is why Section 9’s dual-scale bound (Section 9.4) is stated over $\mathbb{R}$ rather than only instantiated on integers.

10.2 Reproducibility

Every theorem cited by name in this paper was re-verified at the commit given in Appendix B using `#print axioms`:

```
lake env lean -stdin < axiom_audit_script.lean
```

run from `papers/publication/`; the full output is in `axiom_audit_log.txt` in the same directory, and reproduced in Appendix B. Every declaration depends on at most the three standard Lean axioms; none depends on `sorryAx` or on any custom axiom. This particular script and log cover the five-library Mathlib-free core; the `StringTheoryFormalization` and `DualScaleStream2` figures quoted in Section 9 (326 and 99 theorems respectively) are a separate, aggregate audit run via `tools/axiom_audit.py` and are quoted here only at that aggregate level, not reproduced declaration-by-declaration in Appendix B. A companion Appendix A gives, for every headline claim in this paper, the exact Lean identifier, its type (informally), its tier, and a one-line adequacy note — the same table used to produce the corrections in this revision, so that a reader can re-run the audit rather than trust it.

The dependency DAG of the corpus (generated by `tools/leangraph_corpus_analyzer.py`; data in `graph/leangraph.json`) reports 538 nodes and 807 edges, verified acyclic (`is_dag =`

`True`), with 13 redundant transitive edges identified for Hasse-diagram pruning. (An earlier revision of this paper, and the separate internal `PEER_REVIEW_REPORT.md`, quote different node/edge counts, 528/785 and 422/554 respectively; these are stale snapshots from different points in the corpus’s history rather than three independent measurements of the same state. We recommend regenerating this figure immediately before each release rather than carrying it forward across revisions.)

10.3 Related formalized-physics work

This is a small, narrowly-scoped artifact relative to general-purpose physics formalization libraries. `PhysLean/HepLean` (Tooby-Smith et al.) formalizes index notation and a growing body of high-energy-physics results directly in Lean 4 on top of Mathlib [28]; Mathlib itself contains substantial differential geometry (manifolds, vector bundles, and Riemannian metrics) that the present corpus does not use. A natural next step is re-expressing Theorem 3.2 and Proposition 5.1 in that setting.

10.4 Roadmap: promoting Tier L/C claims toward Tier A

The following sketches are **written but not compiled** (they require adding Mathlib as a dependency, which this repository currently does not have); we present them as a concrete, falsifiable roadmap rather than claim them as verified.

```
import Mathlib

/-- Real AM-GM form of Theorem 3.2, with equality iff self-dual. -/
theorem reff_lower_bound {a R : Real} (ha : 0 < a) (hR : 0 < R) :
  2 * Real.sqrt a <= R + a / R := by
  have hs : Real.sqrt a ^ 2 = a := Real.sq_sqrt ha.le
  nlinarith [sq_nonneg (R - Real.sqrt a), hs, Real.sqrt_nonneg a,
    mul_pos hR hR]

/-- Real form of Proposition 5.1, matching the Int version already proved. -/
theorem toy_potential_zero_iff (phi phi0 : Real) :
  (phi - phi0) ^ 2 = 0 <-> phi = phi0 := by
  simp [sub_eq_zero]
```

Promoting Conjecture 7.1 to any tier above C requires, at minimum: (1) a concrete flux or brane completion consistent with the $\mathcal{N} = 4$ non-renormalization obstruction of Section 5.1; (2) a derivation, not an assertion, connecting that completion’s potential to the discrete data of Table 1; and (3) a first-principles derivation of at least one of the three relations in Section 8 from that potential.

11 Conclusion

This paper is narrower than its earlier revision claimed. It kernel-certifies (Tier A) a set of exact integer and rational identities — the positivity of a dual-scale bound on integer radii, $K3$ topological arithmetic, the M_{24} elliptic-genus cross-multiplication and its corrected factorization, Golay-code parameters, and a genuinely-unique-minimum toy potential (after fixing a real bug) — and it states, without conflating tiers, which further claims are established literature (Tier L: Double Field Theory, the Atiyah–Singer index theorem, string-gas kinematics) and which are this paper’s own conjectures (Tier C: the zero-free-parameter proposal and its three numerical phenomenology relations, one of which, the dark-energy prediction, is already in tension with DESI DR2). We consider this an improvement, not a retreat: a claim that survives being checked

against its own formal artifact and against current data is worth more than a stronger claim that does not.

A Claim Ledger

§	Module	Theorem	Tier	Adequacy note
3	UseCase1	self_dual_is_global_minimum	A	$\mathbb{N}$ -instance only; real bound is Tier L
3	DualScale	genesis_no_singularity	A	true by construction of PosScale
4	DualScaleTCC	tcc_cosmic_protection_contract	A	conditional on an assumed field, not derived
5	K3Topology	k3_atiyah_singer_dirac_index	A	correct arithmetic instance of a Tier L theorem
5	UseCase1	moduli_potential_zero_iff (new)	A	fixes the Nat-truncation bug (see box in §5.1)
6	UseCase2	bps_cross_multiplication_lock	A	correct; physical interpretation ($N_Q = 4$) is Tier C
6	GolayHolography	golay_holography_master_contract	A	coding theory only; holographic narrative is Tier C
7	(none)		—	no “zero free parameters” theorem exists; restated as Conjecture 7.1, Tier C
8	Observables	tensor_to_scalar_ratio_reduction (rewritten)	A	exact-fraction identity; the value $r = 1/252$ itself is Tier C
8	Observables	observables_windows_master_contract (rewritten)	A	arithmetic-window conjunction; each window is illustrative, not a confidence interval
6	StringDynamics	M24RepDim_sum_sq (new, 2026-09-16)	A	Burnside regression guard on the corrected EOT (A.3) table; Burnside's identity itself is Tier L
6	Moonshine.EOT	first_five_are_irreps, A6_decomposition, A7_decomposition	A	arithmetic cross-check of EOT (1.12), (1.14)–(1.15) against M24RepDim; the M_{24} action itself is Tier L
6	StringDynamics	bps_ratio_reduced (fixed, 2026-09-16)	A	now norm_num; previously native_decide (axiom Lean.ofReduceBool)
9	DualScaleStream2 (19 modules)	99 theorems total, see Section 9	A	aggregate audit (tools/axiom_audit.py); individual claims/tiers itemized in Section 9

B Axiom Audit

Generated from papers/publication/ by:

```
lake env lean -stdin < axiom_audit_script.lean
```

at commit e831efb on toolchain v4.33.1; full output in axiom_audit_log.txt (same directory).

Declaration	Axioms
K3Topology.k3_euler_characteristic	none
K3Topology.k3_atiyah_singer_dirac_index	none
CourantAlgebroid.jacobiator_vector_vanishes	propext, Quot.sound
CourantAlgebroid.strong_section_condition	propext, Quot.sound
UseCase1.self_dual_is_global_minimum	propext, Quot.sound
UseCase1.moduli_vacuum_stability	propext
UseCase1.moduli_potential_nonneg	propext, Quot.sound
UseCase1.moduli_potential_zero_iff	propext, Quot.sound
UseCase2.bps_cross_multiplication_lock	none
UseCase2.bps_ratio_coprime	none
Observables.tensor_to_scalar_ratio_reduction	none
Observables.observables_windows_master_contract	propext, Classical.choice, Quot.sound
SocrateAI.DualScale.genesis_no_singularity	none
SocrateAI.DualScale.buscher_involution	propext, Quot.sound
DualScaleTCC.tcc_cosmic_protection_contract	propext, Quot.sound
GolayHolography.golay_holography_master_contract	none

No declaration in this table depends on `sorryAx` or on any axiom beyond the three standard Lean 4 axioms. A separate search for a bare `sorry/admit` tactic (as opposed to the word appearing in a docstring) across all five libraries returns zero matches.

Corpus-wide aggregate (2026-09-17). The table above is a fixed, per-declaration snapshot of the five-library Mathlib-free core at commit `e831efb`; it is not re-run against the two newer, Mathlib-dependent libraries. Those are audited in aggregate by `tools/axiom_audit.py` (a different tool from the `axiom_audit_script.lean` above, run in the same 2026-09-17 session but not tied to a single quoted commit): **StringTheoryFormalization** (Stream 1’s sixth library) contributes 89 theorems to a six-library Stream 1 total of **326 theorems, 0 failing**, and the new **DualScaleStream2** library (Stream 2, 19 modules) reports **99 theorems, 0 failing**. In both cases every theorem depends on at most the three standard Lean axioms; neither audit found a `sorryAx` or custom-axiom dependency. Section 9 gives the per-module breakdown and itemizes which physical (as opposed to purely mathematical) statements attached to these theorems remain Tier L or Tier C. We do not claim “zero axioms” anywhere in this corpus: the correct, narrower statement — proved with no axiom beyond Lean’s three standard ones — is what all of the figures in this appendix and in Section 9 support, and is the only claim we make.

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